

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

Three-Week Delivery Block 3/3 | 19-25 September 2026

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Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 9
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Historical CodeGate CTF file.xz/file capture supplied for offline analysis

1. Source Materials and Slide Alignment

- Decrypt Wireless Packets: IEEE 802.11 frames, WEP/RC4, IV reuse, Aircrack-ng key recovery, airdecap-ng decryption, file carving and optional passphrase analysis.
- The slide uses the historical CodeGate CTF 2015 Good Crypto capture and the known WEP40 key A4:3D:F6:F3:74.
- The assignment asks for host/server IP addresses, MAC addresses, manually extracted images/HTML and the router password where the optional CTF challenge is completed.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical wireless CTF capture protected by deprecated WEP40 has been supplied. You will preserve and decompress the evidence, identify IEEE 802.11 traffic, recover or validate the 40-bit key using Aircrack-ng, decrypt the capture offline, inventory network endpoints, carve transferred files and explain why WEP is cryptographically unsafe.

3. Learning Outcomes

- Identify management, control and data frames in an 802.11 capture.
- Explain WEP40 structure, RC4 keystream use, IV reuse and integrity limitations.
- Use Aircrack-ng on a supplied historical capture only.
- Decrypt WEP traffic using airdecap-ng and verify output files.
- Extract IP/MAC endpoints, protocols, images and HTML from decrypted traffic.
- Distinguish a WEP key from a human passphrase and document optional CTF password analysis.
- Recommend modern wireless security controls.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Do not capture, crack or connect to any live wireless network. This lab is exclusively offline and limited to the supplied historical training file.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Offline capture analysis.
Wireless tools	aircrack-ng suite	Recover/validate key and decrypt capture.
Packet tools	Wireshark and tshark	Inspect encrypted/decrypted frames.
Carving tools	foremost, binwalk, file, strings	Recover transferred objects.
Compression	xz-utils	Decompress file.xz.
Evidence	Historical CodeGate CTF capture	Authorized offline dataset.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
```

Screenshot required: Compressed evidence, decompressed capture type and SHA-256 hashes.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab9-Samuel_Chinedu_Onu
Trainee name	Samuel Chinedu Onu
Date and time started	25 th Sept 2026, 10am
Evidence file name(s)	File.xz, file_working.xz, file_xz_sha256.txt
Source or generation method	From ICDFA source github repository
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Working-copy SHA-256	c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d

Field	Student Entry
Analysis workstation/VM	ICDFA Ubuntu Practice hub
Notes on any changes	

8. Part A - Inventory the Wireless Capture

```

CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r "$CAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv

```

```

student@student000442:~/SBT-DF203-Lab9$ CAP=working/file_working capinfos "$CAP" | tee reports/capinfos.txt
capinfos.txtstudent000442:~/SBT-DF203-Lab9$ capinfos "$CAP" | tee reports/capinfos.txt
File name: working/file_working
File type: Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 45 k
File size: 14 MB
Data size: 13 MB
Capture duration: 272.969795 seconds
First packet time: 2015-03-06 07:51:54.364525
Last packet time: 2015-03-06 07:56:27.334320
Data byte rate: 49 kBps
Data bit rate: 395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256: c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1: cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order: False
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 45169
tee reports/protocol_hierarchy.txtab9$ tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
=====
Protocol Hierarchy Statistics
Filter:
wlan frames:45169 bytes:13511274
wlan.mgt frames:224 bytes:35674
data frames:15713 bytes:13104128

```

```

=====
Protocol Hierarchy Statistics
Filter:

wlan                                frames:45169 bytes:13511274
  wlan.mgt                          frames:224 bytes:35674
  data                              frames:15713 bytes:13104128
=====
tshark -r "$CAP" -Y 'wlan' -T fields \
-e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
| head -n 100 | tee reports/wlan_frame_sample.tsv
=====
1   Mar 6, 2015 07:51:54.364526000 UTC   1   11
2   Mar 6, 2015 07:51:54.364527000 UTC   1   11
3   Mar 6, 2015 07:51:54.364528000 UTC   1   11
4   Mar 6, 2015 07:51:54.364529000 UTC   1   11
5   Mar 6, 2015 07:51:54.364530000 UTC   1   11
6   Mar 6, 2015 07:51:54.364531000 UTC   1   11
7   Mar 6, 2015 07:51:54.364532000 UTC   1   11
8   Mar 6, 2015 07:51:54.364533000 UTC   1   11
9   Mar 6, 2015 07:51:54.364534000 UTC   1   11
10  Mar 6, 2015 07:51:54.364535000 UTC   1   11
11  Mar 6, 2015 07:51:54.364536000 UTC   1   11
12  Mar 6, 2015 07:51:54.364537000 UTC   1   11
13  Mar 6, 2015 07:51:54.364538000 UTC   1   11
14  Mar 6, 2015 07:51:54.364539000 UTC   1   11
15  Mar 6, 2015 07:51:54.364540000 UTC   1   11
16  Mar 6, 2015 07:51:54.364541000 UTC   1   11
17  Mar 6, 2015 07:51:54.364542000 UTC   1   11
18  Mar 6, 2015 07:51:54.364543000 UTC   1   11
19  Mar 6, 2015 07:51:54.364544000 UTC   1   11
20  Mar 6, 2015 07:51:54.364545000 UTC   1   11
21  Mar 6, 2015 07:51:54.364546000 UTC   1   11
22  Mar 6, 2015 07:51:54.364547000 UTC   1   11

```

Frame Category	Type Value	Purpose	Example Evidence
Management	0	Association, authentication, beacons	
Control	1	Flow control and acknowledgements	
Data	2	Carries network payload	

9. Part B - Identify WEP Protection and Key Parameters

1. Confirm that protected data frames are present.
2. Record BSSID, source/destination MAC addresses and IV-related fields where dissected.
3. Explain that 64-bit WEP normally combines a 24-bit IV with a 40-bit secret key.
4. Explain why repeated IVs and RC4 weaknesses make WEP unsuitable.

```

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
| head -n 200 | tee reports/wep_protected_frames.tsv

```

```

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt

```

```

99 Mar 6, 2015 07:51:54.382446000 tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
| head -n 200 | tee reports/wep_protected_frames.tsv -Y 'wlan.fc.protected==1' -T fields \
9432 1425628319.232982000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x67ffec 0
10130 1425628320.138252000 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 00:26:66:55:97:d6 0xc2cbeb 0
10155 1425628320.176656000 f0:f6:1c:68:96:7c 04:8d:38:48:e5:b4 00:26:66:55:97:d6 0xc3cbeb 0
10157 1425628320.177168000 f0:f6:1c:68:96:7c 04:8d:38:48:a8:b5 00:26:66:55:97:d6 0xc4cbeb 0
10159 1425628320.177168000 f0:f6:1c:68:96:7c 08:10:77:92:7c:2f 00:26:66:55:97:d6 0xc5cbeb 0
10161 1425628320.177168000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc6cbeb 0
10163 1425628320.177168000 f0:f6:1c:68:96:7c c8:3a:35:58:c4:ba 00:26:66:55:97:d6 0xc7cbeb 0
10165 1425628320.177167000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xc8cbeb 0
10167 1425628320.177686000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x68ffec 0
10170 1425628320.180760000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x69ffec 0
10171 1425628320.182288000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10172 1425628320.187417000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6affec 0
10173 1425628320.187408000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10175 1425628320.187919000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcacbeb 0
10177 1425628320.188952000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0x6bffec 0
10178 1425628320.189976000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6cffec 0
10183 1425628320.192016000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcbcbeb 0
10185 1425628320.193048000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6dffec 0
10224 1425628320.259088000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcccbeb 0
10228 1425628320.262680000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6effec 0
10229 1425628320.262678000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x6ffffec 0
10233 1425628320.263184000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xcdcbeb 0
10261 1425628320.419350000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xcecbeb 0
10266 1425628320.420374000 f0:f6:1c:68:96:7c 01:00:5e:00:00:02 00:26:66:55:97:d6 0xcfcbeb 0
10268 1425628320.420374000 f0:f6:1c:68:96:7c 01:00:5e:00:00:fb 00:26:66:55:97:d6 0xd0cbeb 0
10270 1425628320.420886000 f0:f6:1c:68:96:7c 01:00:5e:00:00:02 00:26:66:55:97:d6 0x70ffec 0
10272 1425628320.422421000 f0:f6:1c:68:96:7c 01:00:5e:00:00:fb 00:26:66:55:97:d6 0x71ffec 0
10353 1425628320.583190000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x72ffec 0
10355 1425628320.583700000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x73ffec 0

```

```

11462 1425628340.819734000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcdfec 0
11464 1425628340.819734000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcffec 0
11466 1425628340.820243000 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcffec 0
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
7 0x42e8eb
7 0x28d3eb
6 0x95dfeb
5 0x9415ed
5 0x86e5eb
5 0x7ae5eb
5 0x28e7eb
5 0x19e5eb
4 0xd5d7eb
4 0xd3e8eb
4 0xc915ed
4 0xc7d8eb
4 0xade4eb
4 0x73e5eb
4 0x65d8eb
4 0x4ad5eb
4 0x31e7eb
4 0x2723ed
4 0x10e7eb
3 0xffe5eb
3 0xf609ed
3 0xa307ed
3 0x97dbeb
3 0x96d1eb

```

10. Part C - Recover or Validate the WEP40 Key

Run Aircrack-ng against the supplied capture. Record selected network/BSSID, number of IVs, key length and recovered key. The slide-provided key is A4:3D:F6:F3:74, entered without colons for airdecap-ng.

```

aircrack-ng "$CAP" | tee reports/aircrack_output.txt
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt

```

```

11466 1425628340.820243000 f0:f6:1c:ts shark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \ceb 0
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt 0:26:66:55:97:d6 0xcfffec 0

7 0x42e8eb -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
7 0x28d3eb
6 0x95dfef
5 0x9415ed
5 0x86e5eb
5 0x7ae5eb
5 0x28e7eb
5 0x19e5eb
4 0xd5d7eb
4 0xd3e8eb

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB    depth  byte(vote)
0     0/ 1
1     0/ 1
2     0/ 1
3     1/ 10
4     6/ 9

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

student@student000442:~/SBT-DF203-Lab9$

```

```

# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt key from the slide: A4:3D:F6:F3:74

A4:3D:F6:F3:74
student@student000442:~/SBT-DF203-Lab9$

```

```

A4:3D:F6:F3:74
student@student000442:~/SBT-DF203-Lab9$ cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt

Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0
student@student000442:~/SBT-DF203-Lab9/working$

```

Evidence Handling

The recovered key is part of the historical CTF dataset, not authorization to attack any wireless network. Keep it inside the course

submission.

11. Part D - Decrypt the Capture Offline

```
cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt
tee: reports/all_working_file_hashes.txt: No such file or directory
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..cap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txtfiles_inventory.txt

cd: no such file or directory: working
Total number of stations seen          10
Total number of packets read          45169
Total number of WEP data packets      15477
Total number of WPA data packets      0
Number of plaintext data packets      0
Number of decrypted WEP packets       15477
Number of corrupted WEP packets       0
Number of decrypted WPA packets       0
Number of bad TKIP (WPA) packets      0
Number of bad CCMP (WPA) packets      0
  4488855  12540 -rw-r--r--    1 student student  12837660 Sep 25 15:08 working/file_working.xz
  4488856  13904 -rw-r--r--    1 student student  14234002 Sep 25 15:08 working/file_working
  4488866  12488 -rw-r--r--    1 student student  12784501 Sep 25 15:32 working/file_working-dec
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
student@student000442:~/SBT-DF203-Lab9$
```

Identify the file created by airdecap-ng, usually with a -dec suffix. Open it in Wireshark and confirm that IP/TCP/HTTP or other higher-layer protocols are now visible.

12. Part E - Extract Endpoints and Conversations

```
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
  | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
```

```

student@student000442:~/SBT-DF203-Lab9$ DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

Decrypted capture: working/file_working-dec
s,eth | tee reports/ethernet_endpoints.txt | tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
=====
Ethernet Endpoints
Filter:<No Filter>
=====

```

	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
Apple_68:96:7c	15407	12517350	7076	616263	8331	11901087
EFMNetworks_55:97:d4	15020	12244019	8178	11650036	6842	593983
ASUSTekCOMPU_2a:c2:7a	358	281920	223	270546	135	11374
IPv4mcast_fb	42	6648	0	0	42	6648
IPv6mcast_fb	36	7022	0	0	36	7022
Broadcast	31	3386	0	0	31	3386
IPv4mcast_7f:ff:fa	31	11277	0	0	31	11277
IPv6mcast_16	10	920	0	0	10	920
IPv6mcast_02	9	630	0	0	9	630
IPv4mcast_16	3	180	0	0	3	180
IPv4mcast_02	2	92	0	0	2	92
NetisTechnol_48:e5:b4	1	42	0	0	1	42
NetisTechnol_48:a8:b5	1	42	0	0	1	42
08:10:77:92:7c:2f	1	42	0	0	1	42
TendaTechnol_58:c4:ba	1	42	0	0	1	42
IPv6mcast_ff:2a:c2:7a	1	78	0	0	1	78

```

=====
s,ip | tee reports/ip_endpoints.txt | tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
=====
IPv4 Endpoints
Filter:<No Filter>
=====

```

	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
192.168.0.15	15309	12508816	6995	608497	8314	11900319
198.90.20.111	8057	6916594	4446	6637205	3611	279389
199.27.79.193	5694	4825296	3131	4603937	2563	221359
192.168.0.9	329	277160	197	265912	132	11248
173.194.127.141	153	109868	80	102504	73	7364
173.194.127.55	143	89699	71	82709	72	6990
8.8.8.8	81	10351	40	6960	41	3391

```

=====
| tee reports/tcp_conversations.txt | tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
=====
TCP Conversations
Filter:<No Filter>
=====

```

	<-		->		Total		Relative Start	Duration		
	Frames	Bytes	Frames	Bytes	Frames	Bytes				
192.168.0.15:51007	<->	198.90.20.111:80	1257	1885 kB	978	73 kB	2235	1958 kB	113.588792000	45.4035
192.168.0.15:50959	<->	199.27.79.193:80	1209	1816 kB	959	71 kB	2168	1887 kB	21.074236000	34.8633
192.168.0.15:51008	<->	198.90.20.111:80	1159	1739 kB	947	70 kB	2106	1810 kB	113.588792000	46.0179
192.168.0.15:51001	<->	199.27.79.193:80	1145	1725 kB	927	66 kB	2072	1791 kB	63.066042000	26.6088
192.168.0.15:51009	<->	198.90.20.111:80	687	1028 kB	572	44 kB	1259	1072 kB	113.605175000	28.7983
192.168.0.15:51014	<->	198.90.20.111:80	664	991 kB	566	44 kB	1230	1036 kB	114.020474000	25.0538
192.168.0.15:51006	<->	198.90.20.111:80	654	970 kB	520	40 kB	1174	1011 kB	113.578039000	25.4942
192.168.0.15:51000	<->	199.27.79.193:80	375	561 kB	323	25 kB	698	586 kB	58.731642000	4.4552
192.168.0.15:50999	<->	199.27.79.193:80	206	303 kB	180	15 kB	386	319 kB	56.792636000	1.9390
192.168.0.15:51036	<->	192.168.0.9:5000	98	144 kB	70	5705 bytes	168	150 kB	243.325633000	0.1777
192.168.0.15:50958	<->	199.27.79.193:80	73	93 kB	67	12 kB	140	105 kB	20.145401000	14.9846
192.168.0.15:51035	<->	192.168.0.9:5000	80	116 kB	56	4756 bytes	136	121 kB	243.281602000	0.1398
192.168.0.15:50983	<->	173.194.127.141:80	71	99 kB	59	4392 bytes	130	104 kB	27.831547000	31.4188
192.168.0.15:50997	<->	199.27.79.193:80	62	87 kB	37	3826 bytes	99	90 kB	56.126972000	11.0492
192.168.0.15:51034	<->	173.194.127.229:443	23	13 kB	32	9731 bytes	55	22 kB	234.384001000	6.7568
192.168.0.15:51005	<->	198.90.20.111:80	25	22 kB	28	4757 bytes	53	27 kB	113.019956000	2.8111
192.168.0.15:50960	<->	31.13.70.1:443	26	15 kB	25	3151 bytes	51	19 kB	21.329724000	32.8418
192.168.0.15:51028	<->	173.194.127.227:443	18	11 kB	29	7603 bytes	47	18 kB	172.217100000	37.9452
192.168.0.15:51003	<->	198.41.208.140:80	21	25 kB	23	2718 bytes	44	28 kB	102.780857000	32.8878
192.168.0.15:51018	<->	173.194.127.55:80	20	24 kB	16	1494 bytes	36	26 kB	115.102391000	30.4286
192.168.0.15:50973	<->	199.16.158.186:443	15	5156 bytes	19	2640 bytes	34	7796 bytes	25.665085000	31.5233
192.168.0.15:51002	<->	104.74.158.12:80	18	20 kB	15	1236 bytes	33	21 kB	63.175099000	30.4746
192.168.0.15:50974	<->	199.16.158.186:443	14	5082 bytes	18	2574 bytes	32	7656 bytes	25.665085000	31.5223
192.168.0.15:51016	<->	173.194.127.55:80	13	15 kB	19	1749 bytes	32	16 kB	115.101879000	30.4439
192.168.0.15:51020	<->	74.125.203.156:80	16	17 kB	16	2534 bytes	32	20 kB	115.146935000	31.5115
192.168.0.15:50956	<->	198.41.209.137:80	13	14 kB	15	1556 bytes	28	16 kB	13.669693000	31.2704
192.168.0.15:51032	<->	173.194.127.231:443	11	5123 bytes	16	1409 bytes	27	6532 bytes	210.917057000	0.2191

```

192.168.0.15:51030 <-> 172.172.239.127:443 0 0 bytes 1 78 bytes 1 78 bytes 210.553984000 0.0000
=====
student@student000442:~/SBT-DF203-Lab9$ tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
22 192.168.0.15 224.0.0.2 IGMPv2
23 192.168.0.15 224.0.0.251 IGMPv2
24 192.168.0.15 224.0.0.2 IGMPv2
25 192.168.0.15 224.0.0.251 IGMPv2
26 8.8.8.8 192.168.0.15 DNS
27 8.8.8.8 192.168.0.15 DNS
30 192.168.0.15 17.172.239.31 TCP
31 192.168.0.15 17.172.239.127 TCP
33 17.172.239.31 192.168.0.15 TCP
35 192.168.0.15 17.172.239.31 TCP
36 192.168.0.15 17.172.239.31 TLSv1
37 17.172.239.127 192.168.0.15 TCP
38 192.168.0.15 17.172.239.127 TCP
42 17.172.239.31 192.168.0.15 TCP
43 17.172.239.31 192.168.0.15 TLSv1

```

13. Part F - Extract Images, HTML and Other Objects

5. First try Wireshark File > Export Objects for HTTP or other recognized protocols.
6. Use tshark object export for HTTP where available.
7. Use foremost as a carving method and document that carved files may include false positives or lack original filenames.
8. Open recovered objects only inside the lab VM and hash them before viewing.

```

mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

```

```

student@student000442:~/SBT-DF203-Lab9$ # Generic carving/http_objects exported/foremost
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

Processing: working/file_working-dec
[*]
exported/http_objects/hls%3fversion=4.12: XML 1.0 document, ASCII text
exported/http_objects/analytics%3fconly=true&visitor_id=73685114&pi_opt_in=&campaign_id=1564&account_id=47382&title=Decoding%20Zeus%20Disguised%20as%20an%20RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer.com%2Fr%2Fnetsec: ASCII text
exported/http_objects/Figure-15.png: PNG image data, 470 x 145, 8-bit/color RGBA, non-interlaced
exported/http_objects/v2(1): JSON text data
exported/http_objects/b90BgL0iMZfDSpmcXcE8nPZRQubYfRc-1FWT9T9EE8M.woff: Web Open Font Format, TrueType, length 13324, version 1.1
exported/http_objects/_utm.gif%3futmvv=5.6.3dc&utms=1&utm=1196661767&utmhn=phishme.com&utmcs=UTF-8&utmsr=320x568&utmv=320x460&utmcs=32-bit&utm=en-us&utmje=0&utmfl=-&utmdt=Decoding%20Zeus%20.com%7Cutmccn%3D(referral)%7Cutmcmd%3Dreferral%7Cutmct%3D%2Fr%2Fnetsec%3B&utmu=q~: GIF image data, version 89a, 1 x 1
exported/http_objects/jquery-ui.css: ASCII text, with very long lines (551)
exported/http_objects/Figure-4.png: PNG image data, 440 x 173, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-linkedin.png: PNG image data, 87 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/Figure-21.png: PNG image data, 603 x 411, 8-bit/color RGBA, non-interlaced
exported/http_objects/object6681.application%2Fjavascript: JavaScript source, ASCII text, with very long lines (3768)
exported/http_objects/iframe.html: HTML document, ASCII text, with very long lines (2225)
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dfc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22(3).com%2F8iVv7t6%22%7D: ASCII text, with no line terminators
exported/http_objects/lumbar.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dfc88d838afa4c1a3d%22%2C%22accountId%22%3Anull%2C%22sessionId%22%3A%224d0d205eea8f1f11bd5b098028b761dc%22%2C%22platform%22%3A%22web%22%2C%22source%22%3A%22(1).com%2F8iVv7t6%22%7D: ASCII text, with no line terminators

```

```
e81216e67645c4363da811baf86e4b6a68aaa850e43765323ef58b5293e6d78
e1eadea754a617805d616d9faa28c8211a36421c11864d4152826ff03d3dcb
e90760c360ef82aa247fdc2b79ce6cc0e465b119d70aec68e89d7850fe07569
68e18431ffffcd3e5d06bcfce5510c65ac7720b3e667833531c67b8d5ae61d3
f97328c08c8b6e8279e4cba83b95f1faf4e264a07e593c55b3e6753f5d1a0c6
936530c477845cdda0828673cab8f3ad0cf9cf589123a6e9d15413e1fd1a083
330a0196da3072001cf3494d57e3ae7bd7ed26db7214a24e9f7488d2e7e9d54
3b0c44298fc1c149afbfb4c8996fb92427ae41e4649b934ca495991b7852b855
9857c891BBAF714&r=0.1958578866906464
855e8af2188864f04fe1e602db1841b0bc8472680ed943d8fb31dc67c0e3b7a
dd50581af2d4876bb2311961b8fc1116d7d115bdcbe4198b587ee79cf4040bd
9ebace7202b39be46546a909c5ae92c2a34245cba32537b63564c22169e34cf
b330bc7f2e4faab709f44958cb276cfb497ede39138aa0e8a4fbf4fc5a69653
00121e2bd294cf2db4594b660c300b1893590c8e0692ed7bd6fbca6751b0e6d
7e7a49f6d21fbef846aba4bd271502d7ec9489bfb3fd96f5ff7cf19140875e
546b5559b78314d342d18d290751916a34bf5ca20a1b565cdef64274a7a1d59
8a6aead815e4147f2376ee4b902ef7db68cba24dac17b679d8a99e460ef8873
9733bcfae19974b7c76d6426dd3391105876c33dfdc673b1d0f14cc0aa10574
4d24f6b27cc7ceea56fbec786bb1f486fdad9a1f998f760f76d1f44671e105c
a69c1c1a562281b6ae35d95f1e9bba3d6897e18ccf5d6759bf964e232cb3c80
92477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676
bf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a
81fb2c21414bfb0c7cedbbe1d57f9506424033bc89beed026e85ea6c85d7ccf
34c09b9b2d436a30c260eaad82679d2a2c970d4962399a8da227ee07162a755
e1c79d0b533ec5ff0acf199d84fbc7dc7e07acbe9defcac65416b63687950d
df5dd4e0ef2b185e8bb0af7b6e90abe74a84384cb4700658e76e754c8bfe550
d11ac950ff1be55ca0e69a487331563f958afd8295252ca88c96c4e72b161eb
2e357aea1b284a9e6a710885744310191740a96d99bb189146b1cbc547ea79e
f1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629
337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
0d3a0aff7dc3bf32d2176fc3dcda6e7aba2867c4f4d1f7af6355d2cfc6c44f8
3f7e56783c414746a9fc3612637509560424617e3135248640eac142d6e0f94
337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015
de95dfbcc35d61cf435401e653d23fae5c98920d34dbb3622a526e645372b0a
891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc
f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6
tudent@student000442:~/SBT-DF203-Lab9$
```

```
exported/http_objects/object6488
exported/http_objects/share-email.png
exported/http_objects/object6524
exported/http_objects/Figure-5.png
exported/http_objects/Figure-16.png
exported/http_objects/object6689.application%2fjavascript
exported/http_objects/nivo-slider.css
exported/http_objects/admin-ajax.php%3faction=wordfence_logHuman&hid=305E48E6D496430
exported/http_objects/twitter.png
exported/http_objects/redir.html
exported/http_objects/trackalyze.js
exported/http_objects/logo.png
exported/http_objects/qpy-ViLNKP-Vf0dbcs6r6-vvDin1pK8aKtelpZ5c0A.woff
exported/http_objects/bid%3fsrc=3079&u=http%3A%2F%2Fimgur.com%2F8iVy7t6&cb=3908366
exported/http_objects/blah.css
exported/http_objects/sharePlayer.css%3f1425580383
exported/http_objects/homewave-upwhite.png
exported/http_objects/collapsible-widgets.js
exported/http_objects/sharePlayer.js%3f1425580383
exported/foremost/png/00020736.png
exported/foremost/png/00010214.png
exported/foremost/png/00024242.png
exported/foremost/png/00001831.png
exported/foremost/png/00001765.png
exported/foremost/png/00010671.png
exported/foremost/htm/00001639.htm
exported/foremost/htm/00024427.htm
exported/foremost/gif/00002450.gif
exported/foremost/gif/00001260.gif
exported/foremost/gif/00000406.gif
exported/foremost/gif/00010456.gif
exported/foremost/gif/00003719.gif
exported/foremost/gif/00003653.gif
exported/foremost/gif/00010876.gif
exported/foremost/audit.txt
exported/foremost/jpg/00004538.jpg
exported/foremost/jpg/00000104.jpg
```